

Sukhrob Ilyosbekov

Machine Learning Engineer · Computer Vision & LLM Systems

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SUMMARY

I came to machine learning after years of shipping regular software, and I still treat a model like any other production system: it has to be retrained, watched for drift, and explained to the person acting on its output. These days that mostly means clinical data, where a wrong prediction lands on a real patient. On the research side I work in computer vision, mainly medical imaging and models that learn images and text together. I'm comfortable with the math underneath, and I like taking a paper apart until I can make it run.

TECHNICAL SKILLS

Deep Learning & CV: PyTorch, TensorFlow, CNNs, Vision Transformers, DINOv3, EfficientNet, YOLO, contrastive learning, GradCAM++, OpenCV

LLM Systems: Claude/GPT/DeepSeek APIs, tool-use agents, MCP, RAG, vision-language models, Hugging Face

Math & Modeling: Probability and statistics, linear algebra, optimization, time-series forecasting, feature engineering, A/B testing, error analysis

Languages: Python, C#, C++, TypeScript, Kotlin

Data & Serving: NumPy, pandas, scikit-learn, FastAPI, PostgreSQL, SQL Server, MongoDB, Redis, OHIF/DICOM

Infrastructure: CUDA, mixed-precision and multi-GPU training, model serving and drift monitoring, Docker, Kubernetes, GitHub Actions, AWS, Azure

Software: ASP.NET Core, SignalR, Blazor, Node.js, Bun, NestJS, React, Next.js, Angular

EDUCATION

Northeastern University, M.S. in Computer Science, GPA 4.0, Boston, MA

Jan 2024 – May 2026

Suffolk University, B.S. in Computer Science, Cum Laude, Boston, MA

Sep 2021 – May 2023

Honors: National Physics Olympiad, 1st Place (Uzbekistan) · Associate Member, Sigma Xi · Coding Competition Winner, TUIT

PROFESSIONAL EXPERIENCE

Machine Learning Engineer | [EmTech Care Labs](#) | Portland, ME

Jan 2025 – Present

- Built a model that reads Alzheimer's residents' daily activity and caregiver logs and flags who is at risk of decline, so staff can step in before a crisis.
- Wrote an NLP pipeline that turns free-text clinical notes into structured records of symptoms, medications, and care events. It replaced manual chart review.
- Both models run in production on de-identified data under HIPAA, with nightly retraining, drift alerts, encrypted PHI, and an audit trail on every prediction.
- Also lead engineering for the care platform itself, a TypeScript monorepo on AWS that passed an outside HIPAA readiness review with no critical findings.

Machine Learning Engineer | [AllFactors](#) | Santa Clara, CA

Oct 2021 – Dec 2024

- Built the property-valuation and market-forecast models that replaced the company's rules-based system.
- Wrote the real-time feature pipeline behind them on SignalR and SQL Server change tracking. New transactions reached the models within seconds and fed a nightly retrain.
- Users saw predictions in Blazor WebAssembly dashboards; automated tests covered the whole path from incoming data to prediction.

Earlier: Software Engineer at Smart Meal Service, Russia (2020 – 2021), writing C# software for food kiosks and a robotic cashier. Game Developer at Pentallight Technology, Malaysia (2020 – 2021), on networking and hand tracking for a Unity VR simulation. Freelance Python NLP and .NET work (2019 – 2020). Game Developer at EC Dev Team, Uzbekistan (2016 – 2019).

SELECTED AI/ML PROJECTS

MorphoCLIP

[GitHub](#) | [Paper](#)

PyTorch, DINOv3, BioClinical ModernBERT, contrastive learning, CPJUMP1

- Matches Cell Painting microscopy of cells treated with drugs or genetic perturbations to plain-language descriptions of the treatment, using contrastive image-text training.
- The DINOv3 and BioClinical ModernBERT backbones stay frozen and only the projection heads train, so it fits on one consumer GPU.
- Evaluated on CPJUMP1 (51 plates, 3M+ images). As first author I led design, training, and evaluation.

Localize, Don't Beautify

<https://arxiv.org/abs/2608.02841>

Image-editing APIs, inpainting, ArcFace, prompt engineering

- Commercial image editors asked for one surgical change tend to retouch the whole face. I tested prompt steering, masked compositing, and inpainting as client-side fixes across six editors on 196 preview edits.
- Scored identity preservation with ArcFace and checked whether each edit stayed in its region. Plain masked compositing kept edits in place best. Sole author.

MelanomaNet

[GitHub](#) | [Paper](#)

PyTorch, EfficientNet V2, focal loss, GradCAM++, OpenCV, ISIC 2019

- Nine-class skin-lesion classifier on ISIC 2019 (25K images). EfficientNet V2 with focal loss to handle the class imbalance; 0.86 weighted F1.
- Scores the model's GradCAM++ attention against the ABCDE criteria dermatologists use, each measured from the lesion mask. Sole author.

Med Image Scanner

<https://github.com/suxrobgm/med-image-scanner>

FastAPI, PyTorch, OpenCV, OHIF Viewer, DICOM, Next.js

- Pulls X-ray, CT, and MRI studies from hospital imaging systems over DICOM and runs PyTorch detectors on them, such as pneumonia on chest X-rays and hemorrhage on head CTs.
- Results show up as overlays in OHIF. Studies are de-identified on the fly, and access is role-based and logged.

LogisticsX

<https://logisticsx.app> | [GitHub](#)

Multi-provider LLMs, MCP, .NET 10, EF Core, SignalR, Angular 21, Kotlin Multiplatform, PostgreSQL

- A multi-provider LLM agent with custom tools does the dispatcher's job: it assigns freight to trucks, checks federal hours-of-service rules, and plans multi-stop routes.
- I built the rest of the product as well: a multi-tenant .NET backend, Angular portals, a Kotlin Multiplatform driver app, and load-board and telematics integrations.

Bookshelf Scanner

<https://github.com/suxrobgm/bookshelf-scanner>

YOLO, Moondream2 VLM, llama.cpp, FastAPI, Angular

- Photograph a shelf and get a book list back. YOLO segments each spine, Moondream2 reads the title and author, and an Angular UI handles corrections and export. Won an Outstanding Project Award at Northeastern.

OGStack

<https://ogstack.dev> | [GitHub](#)

Next.js 16, ElysiaJS, Bun, Satori, Cloudflare R2, Stripe

- Uses Claude, GPT, and DeepSeek to read a site's branding and generate its link-preview image. Served 10K+ images from a global CDN during beta.